Overview of Cybersecurity Importance of cybersecurity in modern enterprises. Key terms and concepts: Cybersecurity, threats, vulnerabilities, and risks. Practical :Explore cybersecurity dashboards and tools. Demonstration of a basic simulated phishing attack setup using email templates.

Tasks

1. Create a table listing 5 key cybersecurity terms (such as threat, vulnerability, risk, asset, attack vector) and write a one-line definition for each in your own words.

Ans. :-

No.	Cybersecurity Term	One-Line Definition
1	Threat	A threat is anything that can potentially cause harm to a computer system, network, or data.
2	Vulnerability	A vulnerability is a weakness in a system that an attacker can exploit.
3	Risk	Risk is the possibility of loss or damage when a threat takes advantage of a vulnerability.
4	Asset	An asset is anything valuable that needs protection, such as data, devices, software, or accounts.
5	Attack Vector	An attack vector is the method or path an attacker uses to gain unauthorized access to a system.

2. Research and find a real-world news article from the last 2 years about a major cyberattack on an Indian company or app (for example, a breach involving Zomato, Paytm, or IRCTC). Summarize what happened and identify one threat, one vulnerability, and one risk mentioned in the incident.

Ans.:-

Real-World Cyberattack: Tata Technologies (India) :-

News article: "Tata Technologies says ransomware attack hit IT assets, investigation ongoing" — TechCrunch, 31 January 2025.

The incident is also confirmed in Tata Technologies' official regulatory filing dated **31 January 2025**.

What happened ? :-

Tata Technologies reported a **ransomware incident** affecting some of its IT assets. As a precaution, some IT services were temporarily suspended and later restored.

Threat :-

Ransomware attackers — malicious actors who attempt to disrupt systems and potentially steal data for extortion. A group called Hunters International later claimed responsibility and alleged that it had stolen about 1.4 TB of data, although this theft claim was not independently confirmed.

Vulnerability :-

The specific vulnerability/root cause was not publicly disclosed by Tata Technologies at the time; the company said an investigation with security experts was underway.

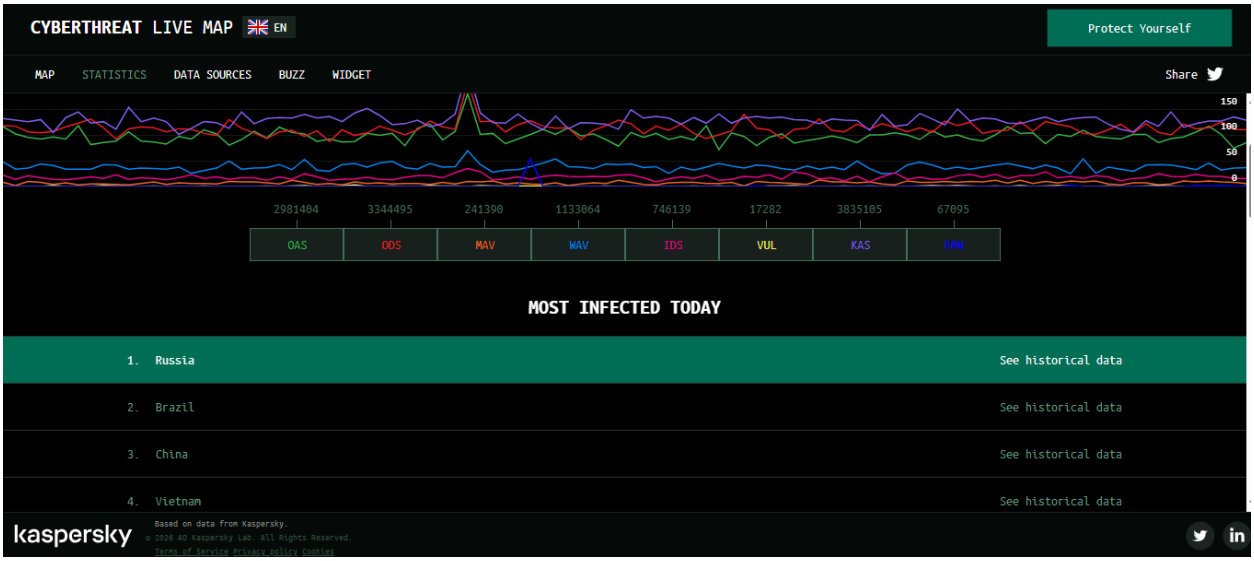
Risk :-

Potential data loss or exposure, service disruption, and financial/reputational damage. The company temporarily suspended some IT services, while client delivery services remained unaffected.

In January 2025, Tata Technologies, an Indian engineering and technology company, experienced a ransomware attack that affected some of its IT assets. The company temporarily suspended certain IT services and restored them afterward. It investigated the root cause with cybersecurity experts. A ransomware group later claimed it had stolen a large amount of data, but that claim was not independently confirmed.

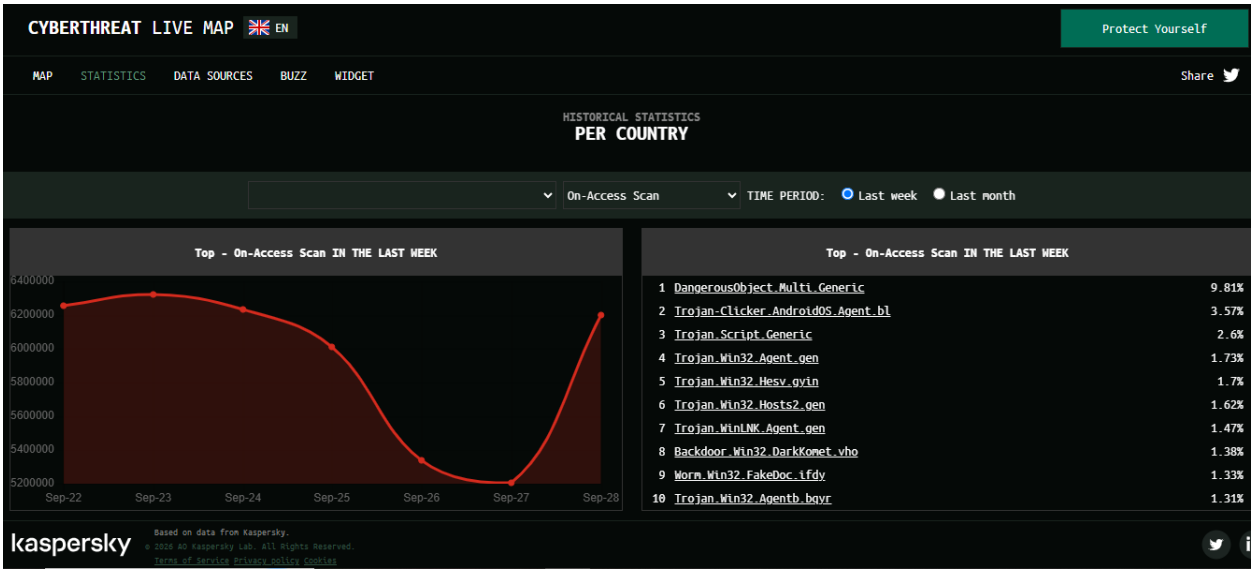
3. Sign up for a free account on any online cybersecurity dashboard or threat intelligence tool (such as https://www.cybermap.kaspersky.com/ or https://threatmap.checkpoint.com/). Take a screenshot of the dashboard and list 3 types of information or metrics you see displayed.

Ans. :-



Steps to complete the practical :-

1. Open the Kaspersky Cyberthreat Live Map.
2. If the site asks for an account, create/sign in to a free account.
3. Open the MAP or STATISTICS section.
4. Take a screenshot showing the dashboard.
5. Submit the screenshot with the following three observations.





3 types of information/metrics :-

No.	Information / Metric	What it shows
1	Detections per second	Shows the number/rate of cyberthreat detections occurring over time.
2	Most infected countries	Shows countries with high numbers of detected threats.
3	Threat categories	Shows different types of detected activity, such as malware, web threats, spam, vulnerability detections, and intrusion-related activity.

I used the Kaspersky Cyberthreat Live Map to observe global cyberthreat activity. The dashboard displays information such as detections per second, the most infected countries, and different categories of cyberthreats.

**4. Write a simple phishing email template targeting users of a popular app (for example, 'Your Paytm account is suspended, click here to verify'). Highlight the suspicious elements that make it a phishing attempt.

Hint: Think about fake links, urgent language, or requests for sensitive info.**

Ans.:-

From: security@payment-verify-example.com
To: user@example.com

Dear Customer,

URGENT: Your Paytm account has been suspended!

We detected unusual activity on your account. You must verify your account within 24 hours, or your account will remain permanently blocked.

[CLICK HERE TO VERIFY YOUR ACCOUNT]
https://paytm-security-example.com/verify

Please enter your password, OTP, card number, and CVV to complete verification.

If you do not complete verification, your account may be permanently suspended.

Thank you,
Paytm Security Team

5. Use ChatGPT or another AI tool to generate a simulated phishing email for a fictional food delivery app (like 'FoodiePal'). Copy the AI-generated email and explain which parts are likely to trick users and why.

Ans.:-

From: FoodiePal Support support@foodiepal-example.test

Dear Customer,

We noticed a problem with your recent FoodiePal order. Your ₹500 refund is currently on hold.

To receive your refund, verify your account within 24 hours by clicking the link below:

[VERIFY REFUND NOW]
https://foodiepal-example.test/refund

You may be asked to confirm your login password and payment information.

If you do not verify your account within 24 hours, your refund request may be cancelled.

Thank you,
FoodiePal Customer Support

Why might this trick users ? :-

Suspicious part	Why it may trick users
"₹500 refund"	A promised refund can make the user curious and encourage them to click.
"within 24 hours"	Creates urgency and reduces the chance that the user carefully checks the message.
"VERIFY REFUND NOW"	A prominent call-to-action encourages immediate clicking.
Fake website	foodiepal-example.test is a fictional domain and should not be trusted as an official service.

Suspicious part	Why it may trick users
Password/payment information request	Asking for sensitive information is a major phishing warning sign.
Fear of losing the refund	Saying the refund may be cancelled creates pressure to act quickly.